

First-principles inputs for thickness-dependent TCAD of ultrathin IWO thin-film transistors

Mahmoud Elrasheedy PI: Chao-Hsin Wu Mentor: Mukul Kumar Status report, 1 October 2026

1. Aim and method

The Silvaco ATLAS model of W-doped In_2O_3 (IWO) TFTs (channel 2.0/6.3/13.2/31.8 nm) uses pure- In_2O_3 DFT proxies for the confinement shift $\Delta E_g(r)$ and mass increase $\Delta m^*(r)$ [1]. These are recomputed here, consistently and specifically for IWO. Quantum ESPRESSO [2], PBE, PAW (pslibrary 1.0.0 [3]), 71/568 Ry; bixbyite Ia $\bar{3}$ [4]; slabs built with the recipe of Lin et al. (p. 21542): In layer removed, one H per surface O, 25 Å vacuum, $3\times 3\times 1$ k -points. All acceptance criteria were fixed before the results; every deviation is recorded. Values are *computed*, not *validated*.

2. Results

Quantity	This work	Reference / check
Bulk lattice constant (PBE)	$a = 10.306 \text{ Å}$ (+1.87 %)	exp. 10.117 Å [4]; Lin 10.30 Å
Bulk gap / CB mass (PBE)	0.889 eV / $0.159 m_0$ ($\alpha \approx 0.6 \text{ eV}^{-1}$)	Lin 0.94 eV / $0.17 m_0$
Vacuum convergence (1 nm slab)	$\leq 0.1 \text{ meV}$ change, 15–35 Å	criterion 10 meV: passed
Relaxed 1 nm slab (67 BFGS steps)	$\Delta E_g = +0.90 \text{ eV}$, $m^* = 0.281 m_0$ ($\Delta m^* = +0.122 m_0$)	Lin +0.94 eV, $+0.13 m_0$: reproduced
Unrelaxed 1 nm slab	$\Delta E_g = +0.07 \text{ eV}$	relaxation is essential
Band edges, relaxed 1 nm slab	EA 3.93 eV, IP 5.72 eV (PBE)	CPU 7.5 and GPU 7.3.1 runs identical
HSE06 check [6] (SG15 NC [7])	ΔE_g : PBE 0.886, HSE 0.940 eV; ratio 1.06 (1.06–1.11)	criterion <10 %: PBE accepted
W site (80-atom cell, 3.1 % W)	24d preferred by 0.258 eV (k -converged, 0.3 meV)	—
W valence	W–O 1.97–1.98 Å $\Rightarrow$ W⁶⁺	Shannon W ⁶⁺ 1.98 Å [5]
W 5d states (bulk and 1 nm slab)	E_F in W-5d-rich states (~50–56 % W); the W level is resonant 1.2–1.8 eV above the host CB bottom, which is filled up to it (corrected 5 Oct.)	preliminary (PBE; moments 0.6–0.7 μ_B , coarse k)

3. Computing resources

Resource	Amount
CERN HTCondor CPU (weighted)	~8,300 core-h
CERN GPU (A100, H100 NVL)	~85–90 GPU-h
of which productive	~35 GPU-h
Laptop (WSL, 10 MPI)	~190 core-h
EOS storage (QE env, GPU container)	0.34 + 1.6 GB
Largest job (2 nm slab, 176 atoms)	30–100 GB RAM

Speed-ups measured. H100 vs 16 CPU cores: **8.2×**, identical energy ($\Delta E = 5 \times 10^{-7} \text{ Ry}$); norm-conserving HSE06 bulk 37 min on GPU vs no progress in 10 h with PAW. **Laptop equivalent.** ~9,900 useful core-h $\approx$ 6 weeks of 24/7 running (realistically 2–3 months); the 2 nm slab and HSE slab exceed its 15 GB memory and cannot run at all. On CERN: ~5 days. **Losses** (~3,000 core-h, ~50 GPU-h, ~2 days): thread oversubscription, wall-time kill, MPI/InfiniBand error loops, PAW-HSE on host CPU, workflow-rule bugs. All fixed and logged.

4. Workflow

One documented tool (dft_flow.sh): automatic CPU/GPU allocation, CPU-twin/upgrade races, watchdog (memory holds, idle and stuck jobs), EOS checkpoints, automatic continuation and job chaining, keepalive and a live dashboard; tested by fault-injection (canary) jobs, which exposed and fixed three bugs.

5. Implications and next steps

(i) The 1 nm confinement point used in the TCAD model is independently confirmed (+0.90 eV PBE; +0.95–1.00 eV HSE-corrected). (ii) PBE's gap error barely affects the confinement *shift* (ratio 1.06), supporting the slab-minus-bulk approach. (iii) W is a W⁶⁺ donor whose electrons partly occupy W-5d states (PBE), consistent with treating the channel donor density as a fitted, film-specific parameter. **Pending:** relaxed 2 nm slab (BFGS step 28, expected 2 Oct), $\Delta E_c/\Delta E_v$ partition, 2 nm IWO slab (automatic chain), then the ATLAS update.

References

- [1] Lin et al., ACS Nano 16, 21536 (2022).
- [2] P. Giannozzi et al., J. Phys.: Condens. Matter 21, 395502 (2009); 29, 465901 (2017).
- [3] A. Dal Corso, Comput. Mater. Sci. 95, 337 (2014).
- [4] M. Marezio, Acta Crystallogr. 20, 723 (1966).
- [5] R. D. Shannon, Acta Crystallogr. A 32, 751 (1976).
- [6] J. Heyd, G. E. Scuseria, M. Ernzerhof, J. Chem. Phys. 118, 8207 (2003).
- [7] M. Schlipf, F. Gygi, Comput. Phys. Commun. 196, 36 (2015).